



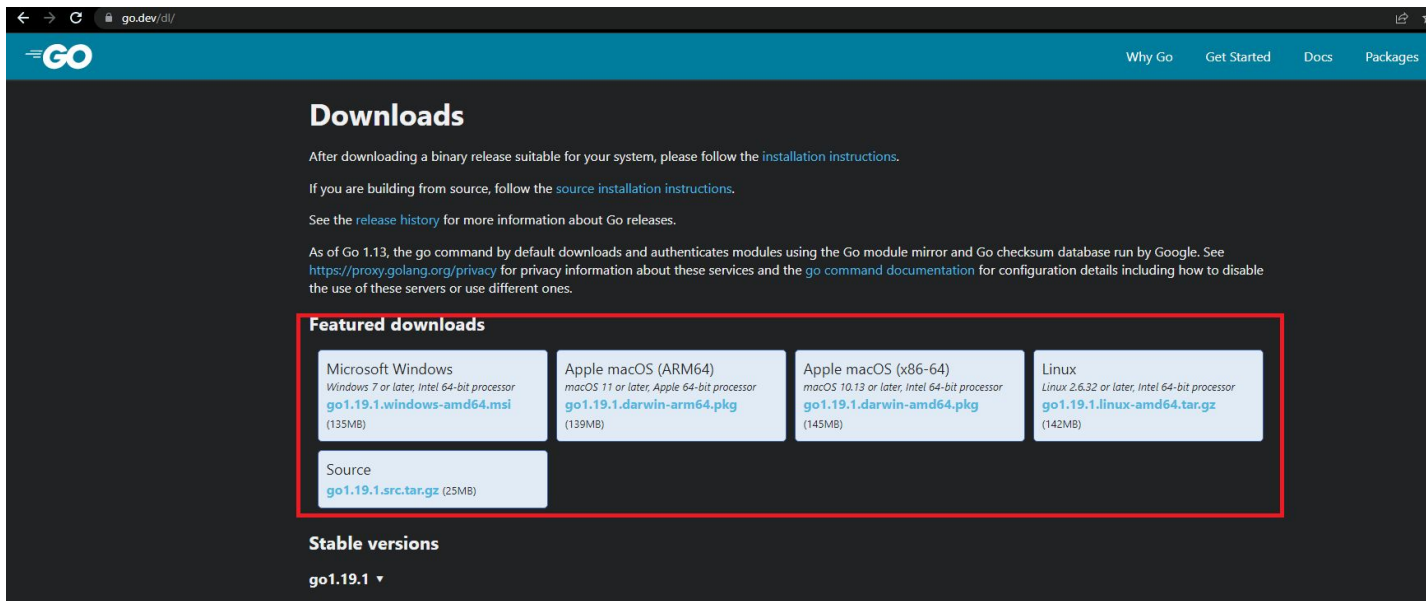
Introducció a

A short horizontal bar with a teal segment on the left and an orange segment on the right.

Què és?

- Llenguatge concurrent, compilat, imperatiu, estructurat i orientat a objectes
- No té excepcions
- Desenvolupat per Google al 2009
- Molt similar a C

Instal·lació



← → go.dev/dl/

GO

Why Go Get Started Docs Packages

Downloads

After downloading a binary release suitable for your system, please follow the [installation instructions](#).

If you are building from source, follow the [source installation instructions](#).

See the [release history](#) for more information about Go releases.

As of Go 1.13, the `go` command by default downloads and authenticates modules using the Go module mirror and Go checksum database run by Google. See <https://proxy.golang.org/privacy> for privacy information about these services and the [go command documentation](#) for configuration details including how to disable the use of these servers or use different ones.

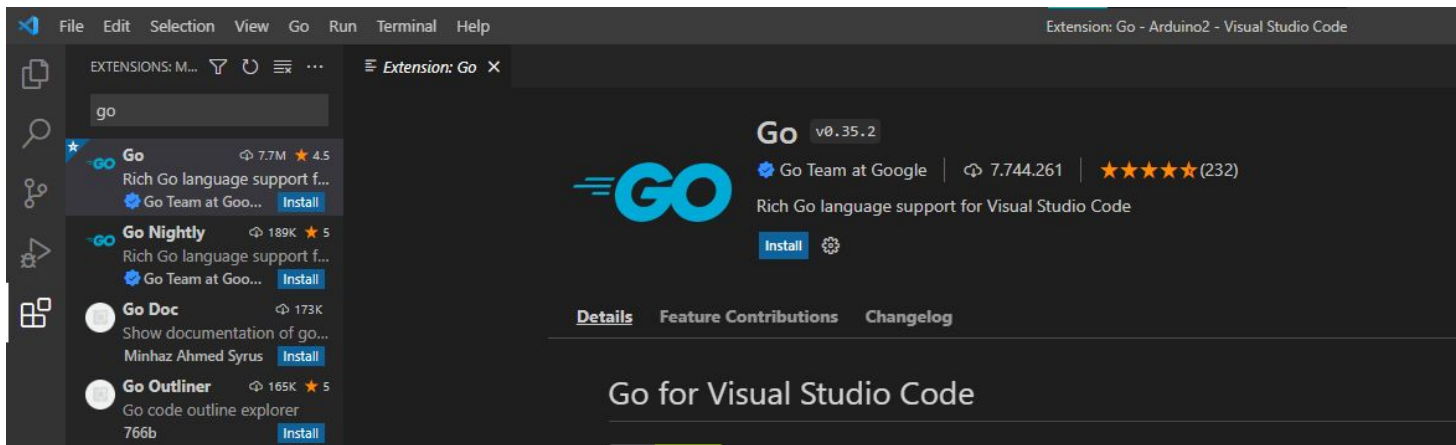
Featured downloads

Microsoft Windows <i>Windows 7 or later, Intel 64-bit processor</i> go1.19.1.windows-amd64.msi (135MB)	Apple macOS (ARM64) <i>macOS 11 or later, Apple 64-bit processor</i> go1.19.1.darwin-arm64.pkg (139MB)	Apple macOS (x86-64) <i>macOS 10.13 or later, Intel 64-bit processor</i> go1.19.1.darwin-amd64.pkg (145MB)	Linux <i>Linux 2.6.32 or later, Intel 64-bit processor</i> go1.19.1.linux-amd64.tar.gz (142MB)
Source go1.19.1.src.tar.gz (25MB)			

Stable versions

go1.19.1 ▾

Instal·lació





Variables

Declaració

```
import "fmt"

var a string

var b, c, d int

func main() {

    fmt.Println("hello world")
}
```



Variables

Assignació

```
import "fmt"

func main() {

    var a int = 30
    var b string = "holaa :D"
    var c float64 = 3.14

    fmt.Println("hello world", a, b, c)
}
```



Variables

Assignació curta

```
import "fmt"

func main() {
    a := 30
    b := "holaa :D"
    c := 3.14

    fmt.Println("hello world", a, b, c)
}
```



Constants

```
import "fmt"

const a int = 10

func main() {

    const b float32 = 20.12
    const c = "holaa :D"

    fmt.Println("hello world", a, b, c)
}
```


A short horizontal bar with a teal left half and an orange right half.

Structs

```
import "fmt"

type Vector struct {
    X int
    Y int
}

func main() {
    var vec Vector

    vec.X = 1
    vec.Y = 0

    fmt.Println(v1)
}
```



Estructura

Els programes es basen
en paquets.

El programa s'executa a partir
de la funció main del paquet
main.



```
1 package main
2
3 import "fmt"
4
5 func main() {
6     fmt.Println("hello world")
7 }
8
```

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Hello World

```
1  package main
2
3  import "fmt"
4
5  ✓ func main() {
6      |   fmt.Println("hello world")
7      | }
8
```

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Conditionals

Sintaxi

```
if initial_statement; condition {  
    //code  
}  
else {  
    //code  
}
```

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Condicionals

Sintaxi

```
if Opcional initial_statement; condition {  
    //code  
}  
else {  
    //code  
}
```



Conditionals

Exemple 1

```
import "fmt"

func main() {
    if x := 10; x == 10 {
        fmt.Println("si")
    } else {
        fmt.Println("no")
    }
}
```



Condicionals

Exemple 2

```
import "fmt"

func main() {
    cond := false
    if cond {
        fmt.Println("si")
    } else {
        fmt.Println("no")
    }
}
```

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Condicionals

Sintaxi

```
switch initial_statement; condition{  
    case expr1:  
    case expr2:  
    case expr3:  
    default:  
  
}
```




Condicionals

Sintaxi

```
switch Opcional initial_statement; condition{  
    case expr1:  
    case expr2:  
    case expr3:  
    default:  
  
}
```



Condicionals

Exemple 3

```
import "fmt"

func main() {
    switch x := 3; x {
    case 1:
        fmt.Println("u")
        break
    case 2:
        fmt.Println("dos")
        break
    case 3:
        fmt.Println("tres")
        break
    }
}
```

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Condicionals

Exemple 4

```
import "fmt"

func main() {
    x := 2

    switch x {
    case 1:
        fmt.Println("u")
        break
    case 2:
        fmt.Println("dos")
        break
    case 3:
        fmt.Println("tres")
        break
    }
}
```



Bucles

Sintaxi

```
for initial_statement; condition; final_statement {
```

```
    //Code
```

```
}
```

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Bucles

Sintaxi

Opcional **Opcional**
for initial_statement; condition; final_statement {

//Code

}



Bucles

Sintaxi

Opcional **Opcional**
for initial_statement; condition; final_statement {

//Code

}



Bucles

Exemple 1

```
import "fmt"

func main() {
    for i := 0; i < 5; i++ {
        fmt.Printf("iteracio :%d\n", i)
    }
}
```



Bucles

Exemple 2

```
import "fmt"

func main() {
    x := 0
    cond := true
    for cond {
        if x < 10 {
            x++
        } else {
            cond = false
        }
        fmt.Printf("Valor: %d\n", x)
    }
}
```




Funcions

```
func (r receiver) func_name (params) (return types){  
    // Code  
}
```



Funcions

```
func (r receiver) func_name (params) (return types){  
    // Code  
}
```



Funcions

Exemple 1

```
type Vector struct {  
    X int  
    Y int  
}  
  
func newVector(coordX int, coordY int) Vector {  
    v := Vector{  
        X: coordX,  
        Y: coordY,  
    }  
    return v  
}  
  
func main() {  
    vec := newVector(1, 0)  
  
    fmt.Println(vec)  
}
```



Funcions

Exemple 2

```
import "fmt"

type Vector struct {
    X int
    Y int
}

func newVector(coordX int, coordY int) Vector {
    v := Vector{
        X: coordX,
        Y: coordY,
    }
    return v
}

func (v Vector) imprimir() {
    fmt.Printf("CoordX = %d CoordY = %d", v.X, v.Y)
}

func main() {
    vec := newVector(1, 0)
    vec.imprimir()
}
```

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Execució

```
$ go run hello-world.go  
hello world
```

```
$ go build hello-world.go  
$ ls  
hello-world    hello-world.go
```

```
$ ./hello-world  
hello world
```